

1.) Introduction to: ~~Fun~~ Foundry - youtube videos on foundries. Explain the safety precautions to be taken in foundries.

Ans:-

★ FOUNDRY PROCESS

The foundry process is a series of steps that involves melting metal and pouring it into molds to create metal castings.

★ PATTERNMAKING

A replica of the desired part is made from wood or metal to create the mold.

★ MELTING

The metal is heated in a furnace until it melts, which can reach temperatures of up to 1600°C .

★ POURING

The molten metal is poured into the mold.

★ COOLING

The mold is cooled and the metal solidifies.

★ REMOVING

The casting is removed from the mold.

* FINISHING

The Casting is subjected to finishing processes like grinding, polishing, heat treating and fabricating.

* INSPECTION AND DELIVERY:

The Casting is inspected and prepared for delivery.

Some safety precautions to take in foundaries:

* Personal protective equipment (PPE):

wear fireproof clothing, gloves, shoes and a helmet. Also, wear protective glasses to protect your eyes from infrared rays. The foundry should also provide a face mask to prevent inhaling harmful fumes.

* Hazard Identification and risk assessment: perform a hazard identification and risk assessment.

* Training:

Ensure that all workers receive ongoing training.



★ Machinery maintenance:
Inspect and maintain machinery regularly.

★ Ventilation.
Ensure proper ventilation.

★ Keep areas dry
Keep all areas on the foundry floor
free of moisture.

★ Avoid Synthetic clothing:
Do not wear any synthetic clothing.
Including inner garments.

★ Avoid liquids:
Keep liquids away.

★ preheat metals:
preheat metals.

★ Keep people away:
Keep people away from the area.

★ Avoid sudden movements:
Avoid lifting materials from the floor
or while seated. Use handling aids
and perform lifts in areas with
adequate footing, space, and lighting.

* Equipment Operation:

Do not use any equipment unless you understand its operation. Use all guards and safety devices on equipments.

* Unplug equipment:

Unplug electric equipment before disassembling or cleaning.

2> Explain the need of a pattern.

Types of pattern - solid or single piece pattern, split pattern, loose piece pattern, Match plate pattern, Crated pattern, Sweep-pattern, Skeleton pattern.

3> Allowances -

Materials used for pattern

Ans: - # PATTERN

In Casting process, pattern is the replica of the device, which is the output of Casting process. This when molded in sand forms mold. After filling mold with the molten metal there is a formation of casting. Patterns play a very important role in Casting as they decide the quality as well as perfection in a particular Casting process. The patterns may be made of metals like aluminium, brass, plaster and wax.



★ Functions of patterns.

- A pattern prepares a mould cavity for the purpose of making a casting.
- A pattern may contain projections known as core points if the casting requires a core and need to be made hollow.
- Risers, runners and gates may form a part of the pattern.
- patterns properly made and having finish and smooth surface reduce casting defects.
- properly constructed patterns minimize overall cost of the casting.

★ Materials used for making patterns

- 1.) Wood
- 2.) Metals
- 3.) Plaster
- 4.) Plastic
- 5.) Wax

1) WOOD

- The wood is the most popular material used for pattern making due to its low cost and easy availability.

★ Advantages

- It can be cut in different shapes from normal to difficult.
- It's very light weight.
- Good surface finish can be achieved by planing or sand landing.
- It's easy to work and quickly available.
- It can be preserved by coating of shellac - varnish.

★ Disadvantages

- It's affected by moisture easily.
- It's damaged quickly after some use due to sand and water absorption.

★ Commonly used wood:-

- 1) Mahagoni
- 2) Teak
- 3) Sal
- 4) Shishu
- 5) pine
- 6) Deodar, etc.



2) METALS

- Metals patterns are mainly used where dimensional accuracy is prime priority metal pattern in industry used for large scale production.

★ Advantages

- Metals are used as patterns materials where the number of casting is very high.
- Greater accuracy in dimension can be achievable.
- It has much longer life than wooden pattern.
- It does not affected by moisture and sand abrasion compared to wooden pattern.

★ Disadvantage

- metal's pattern are costly.
- metal patterns are most costlier than wood.
- To get complicated shape and surface finish machining work is required that is at more cost of the pattern.
- It is heavy weight it is heavier than wooden pattern.
- Most of pattern materials gets rusted due to oxidation takes place.

★ Commonly used metals:-

1. Cast iron,
2. Brass,
3. Aluminium,
4. White metal

3) PLASTER

- plaster is mainly used in small job production where product shape is complicated.

★ Advantage-

- plaster is mainly used for small complicated pattern.
- it is used to make core boxes which has intricate shape and needed highest accuracy.
- it has high compressive strength.
- It can be easily cast into desired shape.

★ Disadvantage.

- It should be handled carefully otherwise high chance of breaking.
- It is made of gypsum cement commonly called plaster of paris.

4) PLASTIC

- Resistance of corrosion.
- Less weight
- High strength
- Good surface finish
- high resistance to wear, low shrinkage
- It's made by thermoplastic setting plastic by using mould made by plaster of paris



5) WAX

- This pattern are mainly used in investment casting for the production of single piece casting. pattern always made from two halves. After mold prepared mold is inserted then heat applied to drain out wax.

★ Types of patterns

- 1) Single piece pattern
- 2) Two piece pattern
- 3) Crated pattern
- 4) Multi-piece pattern
- 5) Match plate pattern
- 6) Skeleton pattern
- 7) Sweep pattern
- 8) Lose piece pattern
- 9) Follow board pattern
- 10) Cope and drag pattern
- 11) Segmental pattern
- 12) Shell pattern.

Types of Pattern



Single Piece



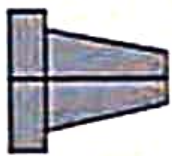
Two Piece



Multi Piece



Loose Piece



Cope and Drag



Sweep



Shell



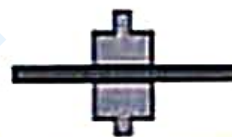
Skeleton



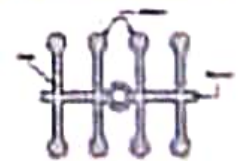
Segmental



Follow Board



Match plate



Gated

1.) Solid or Single piece pattern.

- Single piece pattern is the cheapest pattern among all other types of pattern.
- This pattern generally used in simple processes.
- It is applied into small scale production.
- It is often used for the generation of large castings such as stuffing box of steam engine and for creating simple shapes, flat surfaces like simple rectangular box blocks.
- The important characteristic of this pattern is that there is no need of joint in the mold area.
- In this pattern one surface is considered as flat portion.
- This flat surface is used for parting plane.
- Sand tools are used to cut the sand which ultimately make gating system.
- The modifying becomes a difficult task if there is absence of this flat surface.
- It is expected to lie in cope or drag.



2.) Two-piece pattern.

- Two-piece pattern is also called as split piece pattern. It is popularly used for intricate casting. The shape of casting decides the exact place of parting plane. This parting plane may be flat or irregular surface. In two-piece pattern half part is always molded in drag and other half part is moulded in Cope. The cope part of the pattern has dowel pins. These dowel pins are used to align the two halves of split piece pattern. Holes in the drag half of the two-piece pattern match exactly with dowel pins. It is used in applications where it is very difficult to withdraw casting from the mold. Two piece patterns are used where the depth of casting is very high.

3.) Multi-piece pattern

- Sometimes Castings have very difficult and Complicated design. In such situations multi piece type of pattern are used. 3 or more patterns are included in multi-piece pattern. For instance, if we consider three-piece pattern which comes under multi piece pattern. This three-piece pattern consists of top, bottom and middle parts. This bottom part is drag, top part is cope where the middle part is termed as check box.

4) Match plate pattern.

- Basically match plate pattern is a split pattern. Cope and drag areas are on the opposite faces of metallic plate. This metallic plate is termed as match plate. This type of pattern requires very less work load and gives very high output. Because the gates and runners are also on the match plate. This is used in various manufacturing industries. This is very expensive and gives accuracy as well as high yield. This pattern is widely used for casting metals like aluminium.

5) Grated pattern.

- Grated types of patterns are used to make multiple components inside the single mold. Grated pattern is nothing but the pattern consisting one or more patterns. For joining different patterns gates are used. These are loose patterns where gates are runners have already attached. These patterns are very expensive. Due to their high cost they are used for creating small casting. These are small casting further are used in molding machines as well as in mass producing process.



6) Skeleton pattern

Skeleton pattern is used for casting which have simple size and shape. These castings are usually large in size. The only disadvantages of Skeleton types of patterns are - it is applicable for small number of components and it is not cheap. Economically it is not the best pattern. Stickler is used to remove extra sand. These are nothing but frames of wood that highlight the area which is to cast. These patterns also help molder. They are widely used in process of pit or floor moulding.

7) Sweep pattern.

In sweep pattern we make use of wooden board of proper size is to be rotated about one edge to shape the cavity as circular or rotational symmetry sweep pattern is often used when we have to create casting in very short interval of time. Molds of extensive symmetrical casting can be made easily with the help of sweep pattern. Consists of three parts spindle, base and sweep which is wooden board. Spindle is directed in vertical direction and base is attached with sand.

8.) Loose piece pattern

- It is very difficult to remove one piece of solid pattern which is above or below the parting plane having projections from the mold with the help of loose piece types of patterns projections can be made by loose pieces. It requires skilled labor work as well as it is very expensive. There is one disadvantage of this loose piece pattern is that their shifting can be done due to ramming process.

9.) Cope and Drag pattern

- Cope and drag pattern is a split pattern. This pattern has cope and drag on separate plate. Cope and drag pattern has two parts which are separately molded on molding box. After molding parts, these two separate parts are combined to form the entire cavity. Cope and drag pattern is almost like two-piece pattern. This pattern types are used in the production of large castings where the molds are very heavy and unhandy for a user.



10.) Follow Board pattern

- Follow board pattern consists of tool that is a simple wooden board which is used for several reasons. The wooden board is used as a base in follow board pattern for molding process. This pattern is used in processes where casting structures are weak and they may break after the application of force.

11.) Segmental pattern

- It is just a similar to that of Sweep pattern. The working structure of segmental pattern and Sweep pattern is almost similar. For designing required shape or structure of mold they both employ a part of pattern. As the name suggest segmental pattern is in the form of segments and used for modifying molding circular or round structures. In Sweep pattern there is complete rotation but in segmental pattern there is no complete rotation. For creating mold, we can partly to get required output.

12.) Shell pattern

- Shell pattern is specially used for obtaining hollow shaped structures. Along the center the parting process is done. The resultant halves produced after parting are both clamped.

PATTERN ALLOWANCE

A pattern is a replica of casting which is used to make a mold cavity but it has slightly large dimensions. This change in the pattern is due to when the cast solidifies. It shrinks at some limit due to metal shrinkage property at the time of cooling.

So, to compensate for this, a pattern is made a little bigger.

These slight changes in the pattern are known as pattern allowance.

★ Types of pattern allowance

- These are the following types of pattern allowance to be used in casting process.

1.) Shrinkage Allowance

2.) Draft Allowance

3.) Machining Allowance

4.) Deformation or Camber Allowance

5.) Shake or Rapping Allowance.

1.1 Shrinkage Allowance.

- Shrinkage is defined as the reduction during the cooling or solidification process. This is a common property of all materials.
- The magnitude of shrinkage varies from materials to material, but each and every material has to shrink.
- For avoiding this, the pattern is made larger than the required size of the casting product with the help of the shrink rule.
- Then the difference between actual pattern size and required casting size is known as shrinkage allowance.

2.1 Draft Allowance or Taper allowances.

- During removing the pattern from the mold cavity the parallel surface in the direction in which the pattern is withdrawn are slightly damaged and also converted into slightly trapped surface.

To compensate for these changes these parallel surfaces on the pattern are slightly tapered by about 1 to 2 degrees.

By which removal of the pattern from the mold cavity becomes easy and suitable and also does not affect the casting in any way.

These small changes in the surface of the pattern are called draft allowances to protect it from damage.

3) Machine Allowance.

- As we know that the product of the casting process gives a very poor surface finish, so the surface of the final product of casting always is rough. But we required a product that is polished and has a good surface finish. So in order to have a good surface finish, the final product of the casting is machined with the help of lathe machine, milling machine, shaper machine, slotting machine with these processes such as turning, grinding, shaping, drilling to obtain a surface finish. This allowance is added to the basic size of the pattern. It ranges from less than 2 mm to 15 mm depending on the size and material of the finish.

4.) Shake or Rapping Allowance.

- When the pattern is to be removed from the casting, a slight shake is required to remove the pattern from the sand and this will increase the dimension of the casting slightly. To avoid this increasing in the casting the pattern is made slightly smaller than the casting.

These small changes in the dimensions of the pattern in the casting process are called shaking or rapping allowance.

5.) Deformation or Camber Allowance.

- When the metal is in the cooling process stress is developed during the solidifying of this metal due to uneven metal thickness in the casting process.

This stress can ~~create~~ cause deformation or bend in the casting.

To avoid this bending or deformation in castings camber is provided in the opposite direction so that when bending occurs due to uneven thickness of metal, the casting product becomes straight.

These small changes in a pattern to avoid bending in the casting process are called bending or Camber Allowance.

